

Compositional modification of products from Co-Pyrolysis of chicken manure and biomass by shifting carbon distribution from pyrolytic oil to syngas using CO₂

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ABSTRACT

Co-pyrolysis of chicken manure and biomass was investigated in this study. The pyrolysis of individual samples was characterized by thermogravimetric analysis (TGA) under N₂ and CO₂ atmospheres. This demonstrated that the impact of CO₂ content on the physical aspects of pyrolysis such as onset and end temperatures, and residual mass was negligible. However, a high CaCO₃ content (17 wt%) in chicken manure catalyzed the Boudouard reaction. Despite its negligible physical influence, CO₂ evidently affected the co-pyrolysis of chicken manure and biomass chemically. It expedited the thermal cracking of hydrocarbons from the co-pyrolysis of chicken manure and biomass. Moreover, between 550 and 660 °C, CO₂ reacted with condensable hydrocarbons, effectively improving CO generation. This observation suggested that CO₂ acted as both carbon scavenger and oxygen donor in the co-pyrolysis of chicken manure and biomass, a driving force for shifting carbon distribution between pyrogenic products. For example, pyrolytic oil was transformed into syngas, especially CO, offering an innovative means to modify compositions of pyrolytic products. These effects were not observed in the presence of CaCO₃ and/or CaO.

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1. Introduction

Over the last decades, global energy production has increased by 150%, equivalent to 13.7 billion tons of petro-oil [1]. However, the major resources utilized in chemical and energy sectors were unchanged, still relying on fossil fuels for 80% of global energy production [2]. This heavy dependence on fossil resources has triggered severe socio-economic problems like energy insecurity due to limited and unevenly distributed natural resources [3–5]. In addition, it drives the perturbation of natural carbon cycle and harms the environment [6,7]. The massive carbon output from anthropogenic activities is overwhelming our ecosystem. Greenhouse gases and particulate matters from complete and/or incomplete combustion processes have been recognized as a crucial

factor for triggering global warming [8–10]. To cope with socio-economic and environmental issues stated above, renewable and sustainable energy sources as an alternative to fossil-derived fuel has been extensively developed [11,12]. Among these, biofuels are viable options owing to their intrinsic carbon neutrality [13–15]. In addition, their high content of oxygen offers a better oxidation performance, which reduces the generation of air pollutants such as particulate matter. Furthermore, the mandatory use of biofuels has been politically supported by legislated renewable portfolio standard (RPS) and renewable fuel standard (RFS), which expedites the expansion of biofuels [16,17]. Biofuels are advantageous because they can utilize the pre-existing distribution network without engine modification [18,19].

The first generation of biofuel that is converted from edible biomass has been commercialized. However, its expansion resulted in unexpected side effects (i.e., ethical dilemma, crop price increase, and water shortage) [20,21]. To abate these problems, second (i.e., biofuels from inedible biomass) and third generations (i.e. biofuels

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from aquatic biomass) of biofuels have been investigated over the last decade [22,23]. Despite technical advances, the processes for biofuel production has not yet been commercialized due to their technical inadequacies [24]. The insecure supply chain for biomass due to the regional and seasonal restrictions has been regarded as one of the demerits [25,26]. Therefore, alternative biomasses as raw materials to produce biofuels are in demand. Municipal solid waste (MSW) and agricultural waste [27] such as livestock manure are viable options. The establishment of concentrated animal feeding operations (CAFOs) leads to massive generation of livestock wastes and manures [28,29]. Animal wastes are normally used as an effective fertilizer via composting process [30,31]. However, the surplus of livestock manure, which assimilated nutrients like nitrogen, phosphorous and potassium, resulted in unwanted eutrophication that damaged plants by disrupting the absorbance of other nutrients [32]. In response, anaerobic digestion (AD) has been employed to dispose animal wastes [33]. However, biogas from the AD process requires rigorous purification and extensive concentration steps because of its high content of CO_2 (50–60%) [34]. Moreover, one of the crucial steps in the AD process generates sludge. Hence, a method for sludge disposal is required [35]. Thus, biofuel processing technologies to resolve all these demerits are greatly desired.

Thermo-chemical treatments (i.e., pyrolysis and gasification) of manure have potential applications in fuel processing technology, as it could be applied to most carbonaceous materials, insensitive of biomass type [36]. The syngas from the thermo-chemical processes contributes to energy density and can be used as initial feedstock for synthesizing fuels (e.g., ethanol, gasoline, kerosene, diesel, and jet fuel) and chemicals [37,38].

Despite their several technical advantages, the thermo-chemical processes are energy intensive. Thus, a new way to increase their thermal efficiency and modify their products without catalysis is demanded. To further promote sustainability, CO_2 can be used in the thermo-chemical processes. As an intermediate step for gasification [39,40], pyrolysis was studied herein. In addition, there has been limited number of studies about co-pyrolysis of biomass and animal manure. Therefore, we focused on understanding the effect of CO_2 on co-pyrolysis of biomass and chicken manure. First, the thermolysis of chicken manure and biomass in CO_2 was characterized by thermo-gravimetric analysis (TGA). To produce more realistic data, a scaled-up experiment was also conducted using a batch-type tubular reactor (TR). The effluents of N_2 and CO_2 in the TR were compared and characterized to understand the mechanistic role of CO_2 in co-pyrolysis.

2. Experimental

2.1. Materials

Chicken manure and Zelkova wood were obtained from the Rural Development Administration-National Institute of Animal Science (RDA-NIAS) in Jeonju, Republic of Korea. The collected samples were dried at 60°C for 1 d under 20 Torr in a dry oven. Thereafter, they were stored in a plastic bottle. Prior to all experiments, the samples were dried at 60°C for 1 h. For fundamental analysis, the chicken manure sample was washed with 0.25 M hydrochloric acid (HCl) at 400 rpm for 2 h. Subsequently, its pH was adjusted by rinsing with deionized water, and then the sample was dried at 60°C for 1 d. Moreover, 5 M HCl was purchased from SAMCHUN chemical. Ultra high-purity N_2 and CO_2 gases were purchased from Green Gas and Kuk-Je Industrial Gases, respectively.

2.2. Sample characterization

The inorganic elements were characterized using an iCAP 6000 Series ICP Emission Spectrometer (Thermo-Fisher Scientific). The morphology of the chicken manure sample was analyzed using a Bruker D8 advance X-ray diffractometer (XRD) with Cu K α radiation and LynxEye position-sensitive detector. A proximate analysis was performed by following suggested method (Table 1) [41]. TGA experiments were performed at a heating rate of $10^\circ\text{C min}^{-1}$ from 20 to 860°C under separate atmospheres of N_2 and CO_2 using a Mettler Toledo TGA/DSC star system. The sample loading was 10 ± 0.1 mg, and the total flow rate (i.e., for purge and protective gases) was adjusted to 100 mL min^{-1} .

2.3. Pyrolysis experiment with a tubular reactor (TR)

A quartz TR was used to pyrolyze chicken manure, biomass, and chicken manure/biomass (50:50 w/w) under N_2 and under CO_2 . The pyrolysis was conducted from 20 to 780°C at a ramping rate of $10^\circ\text{C min}^{-1}$. The reaction temperature was controlled and monitored using a proportional-integral-derivative (PID) controller equipped with K-type thermo-couples. The TR was 25.4 mm in outer diameter, 22 mm in inner diameter, and 0.6 m in length. Two Ultra-Torr Vacuum fittings (Swagelok SS-4-UT-6-600) were connected to both ends of the TR. A sample of 1.4 ± 0.02 g was loaded in the center of the TR. Mass flow controllers (MFCs, Brooks Instrument) were used to control the gas flow rate of N_2 and CO_2 . The schematic diagram of the reactor is provided in the supporting information (Fig. S1).

2.4. Analysis of pyrolytic products

The pyrolytic gases were quantified using a micro gas chromatograph (GC, INFICON) calibrated with a standard gas mixture (RIGAS). The collected condensable hydrocarbons (tar) were qualitatively identified using a GC/TOF-MS system (Agilent 7890/ALMSCO) equipped with an Agilent HP-5MS ($60 \text{ m} \times 0.25 \text{ mm} \times 0.25 \mu\text{m}$) column. Prior to the analysis, the tars were diluted by 20-fold with dichloromethane (Sigma-Aldrich). All experiments were triplicated to ensure reproducibility.

3. Results and discussion

3.1. Thermolysis of chicken manure under N_2 atmosphere

Representative mass decay thermogram and its thermal degradation rate, depicted as differential thermogram (DTG), were incorporated and plotted in Fig. 1(a) and (b), respectively.

Disregarding moisture vaporization, Fig. 1(a) shows that mass decay in the thermolysis of chicken manure occurred in three parts: between 210 and 505°C , 620 and 740°C , and 760 and 860°C . The first mass change from 210 to 505°C indicates the thermal decomposition of organic mass fraction in chicken manure, which is roughly equivalent to 55 wt%. The second mass loss between 620 and 740°C is difficult to describe by TGA. Thus, the chicken manure sample was washed with HCl. Interestingly, significant amount of

Table 1
Proximate analysis of the chicken manure and biomass samples.

	Chicken manure, [wt. %]	Lignocellulosic biomass, [wt.%]
Moisture	4.35 (± 0.21)	3.34 (± 0.19)
Volatile matter	71.45 (± 1.31)	94.78 (± 2.11)
Fixed carbon	7.83 (± 0.36)	1.46 (± 0.06)
Ash	16.38 (± 0.68)	0.43 (± 0.02)

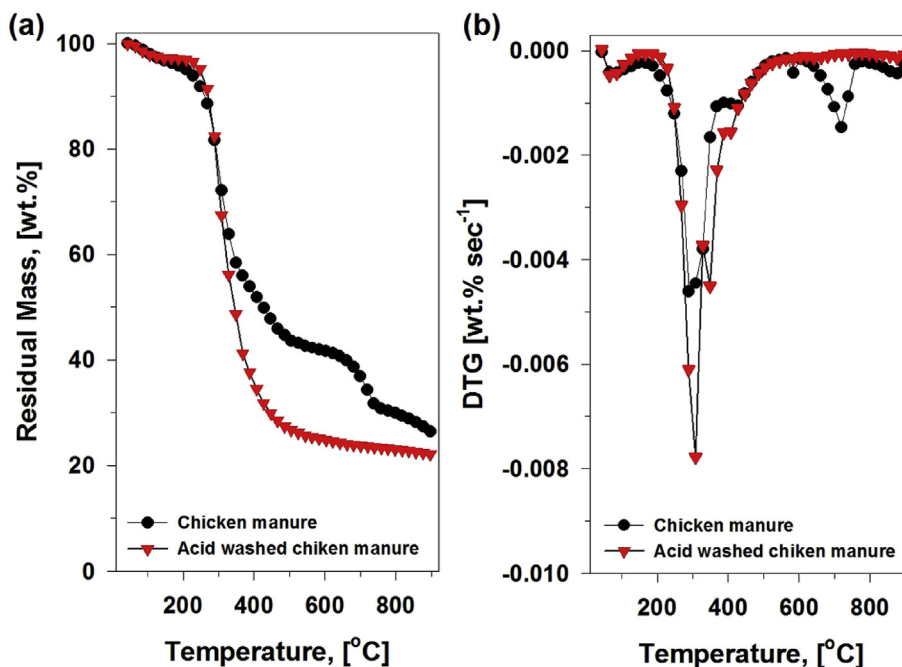


Fig. 1. (a) Representative thermogram and (b) DTG from the TGA of chicken manure (10 ± 0.1 mg) and acid-washed chicken manure (10 ± 0.1 mg) under N_2 (Error bar was not shown due to small deviation of less than 0.1%).

acid-soluble fraction, roughly equivalent to 17 wt% of the original sample mass, was observed. For further studies, the same TGA experiment was conducted for acid-washed chicken manure. The resultant representative thermogram and its thermal degradation rate were compared with the original TGA results. The representative thermogram from the thermolysis of acid-washed chicken manure exhibited only one distinctive mass change, from 210 to 505 °C. This signifies that the second mass loss may be attributed to the thermal degradation of the acid-soluble fraction in the chicken manure sample. To confirm this, the two samples (i.e., chicken manure and acid-washed chicken manure) were analyzed by XRD. Their XRD spectra are depicted in Fig. 2 for comparison.

As shown in Fig. 2, XRD spectra indicates that the soluble mass fraction in chicken manure is mostly $CaCO_3$. Thus, the second mass loss from the thermolysis of chicken manure is mainly attributed to dissociation of $CaCO_3$ into CO_2 gas. To verify this, a TGA experiment for $CaCO_3$ under N_2 was conducted. The resultant representative thermogram and its corresponding DTG were superimposed on

that of chicken manure and is presented in Fig. 3. This suggests that the mass loss from the dissociation of $CaCO_3$ into CO_2 ($CaCO_{3(s)} \leftrightarrow CaO_{(s)} + CO_{2(g)}$), equivalent to 44 wt% of $CaCO_3$ (molecular weight of $CaCO_3 = 100$ and molecular weight of $CO_2 = 44$), initiated at 610 °C and ended at 790 °C. Thus, the temperature regime that demonstrated the second mass loss from the thermolysis of chicken manure represented the dissociation of $CaCO_3$ into CO_2 .

Consequently, the third mass loss in Figs. 1(a) and 3(a) from the thermolysis of chicken manure indicates the occurrence of Boudouard reaction ($C_{(s)} + CO_{2(g)} \leftrightarrow 2CO_{(g)}$), which is thermodynamically favorable above 720 °C. Observably, mass decay continues although the dissociation of $CaCO_3$ into CO_2 ended at 790 °C. Thus, this continuous mass decay after CO_2 depletion suggests that the final mass loss is a consequence of both Boudouard reaction and thermal degradation of refractory organic mass fraction in the chicken manure. Importantly, employing chicken manure as a feedstock for thermo-chemical processes has several advantages over other biomasses. For instance, Boudouard reaction can be initiated by CO_2 from $CaCO_3$. Moreover, catalytic effects can be expected due to CaO from $CaCO_3$. Furthermore, the biochar generated from the pyrolysis of chicken manure can be used as an effective adsorbent for CO_2 , which provides a favorable condition for CO_2 -looping.

3.2. Characterization of the thermolysis of the mixture of chicken manure and biomass in CO_2

To understand the influence of CO_2 on co-pyrolysis of chicken manure and biomass, a series of the TGA experiments was conducted from 20 to 860 °C. Representative thermograms and DTGs from the separate thermolysis of chicken manure, biomass, and the equivalent mass mixture of both under N_2 and under CO_2 were compared in Fig. 4.

In Figs. 1 and 3, three distinctive mass losses in the thermolysis of chicken manure under N_2 were discussed. Interestingly, the thermolysis of chicken manure under CO_2 exhibited two distinctive

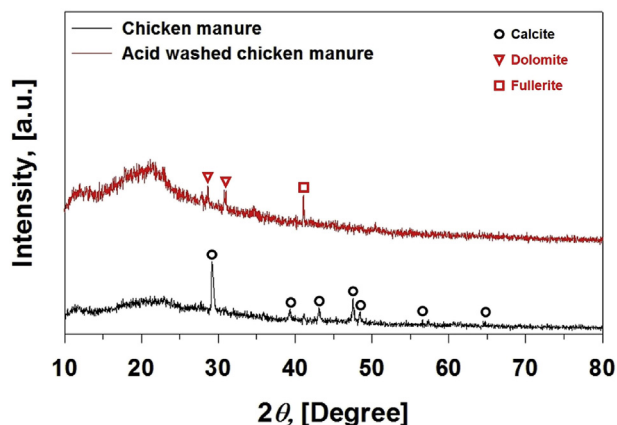


Fig. 2. XRD spectra for chicken manure and acid-washed chicken manure.

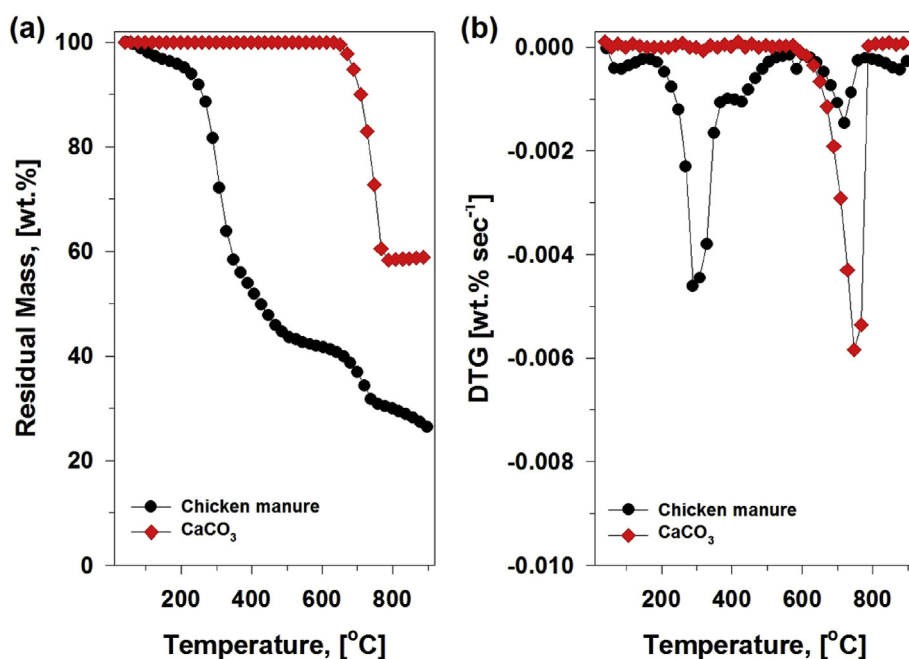


Fig. 3. (a) Representative thermogram and (b) DTG from the TGA of chicken manure (10 ± 0.1 mg) and CaCO₃ (10 ± 0.1 mg) under N₂ (Error bar was not shown due to small deviation of less than 0.1%).

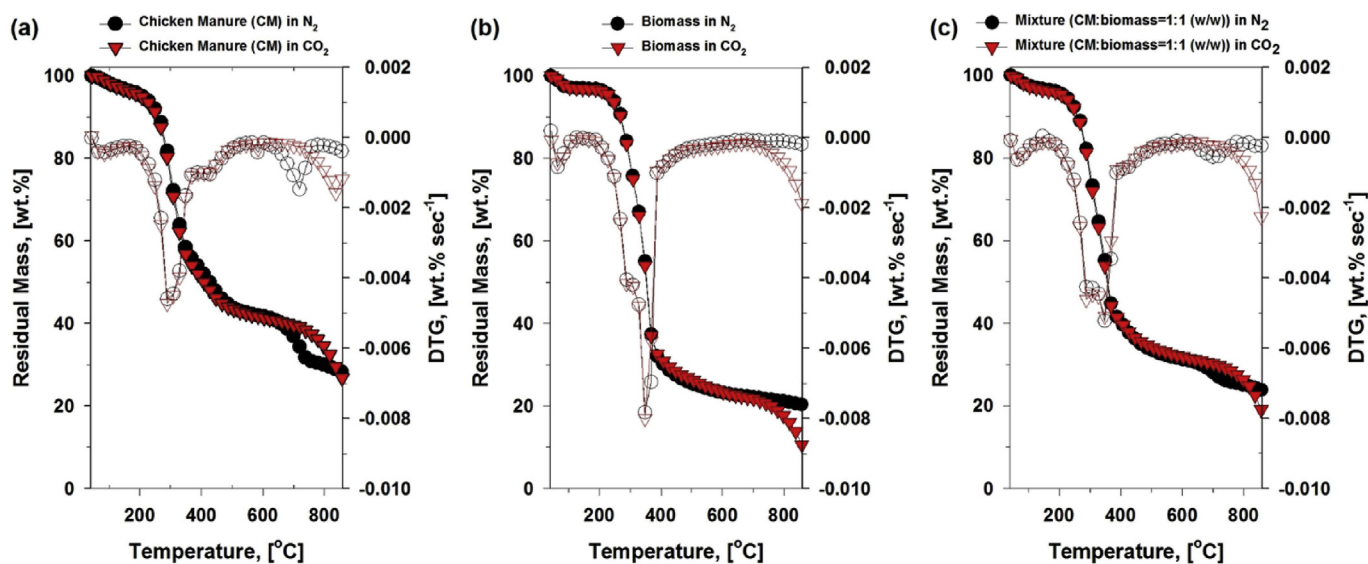


Fig. 4. Representative thermogram and DTG from the TGA of (a) chicken manure (10 ± 0.1 mg), (b) biomass (10 ± 0.1 mg), and (c) the equivalent mass mixture of both (10 ± 0.1 mg) under either N₂ or CO₂ (Error bar was not shown due to small deviation of less than 0.1%).

mass losses. The dissociation of CaCO₃ into CO₂ was not observed under CO₂ atmosphere. This is explained by the high partial pressure of CO₂ under CO₂ atmosphere, which implies the achievement of a dynamic equilibrium. Moreover, the thermal degradation rate of Boudouard reaction (i.e., the third mass loss) under CO₂ is faster than that under N₂. Evidently, the slope for the mass loss due to Boudouard reaction under CO₂ is steeper than that under N₂. The abundance of CO₂ significantly improved the kinetics of Boudouard reaction. Furthermore, catalytic effect could have also expedited Boudouard reaction.

Besides the mass loss due to Boudouard reaction, the thermolysis of corresponding samples under N₂ and CO₂ atmospheres show

comparable thermal degradation patterns. This similarity implies that the influence of CO₂ on pyrolysis is nearly negligible. Therefore, the physical aspects of thermolysis, such as onset and end temperatures, are not affected by CO₂. These experimental observations signify that heterogeneous reactions between CO₂ and sample surface are insignificant as the thermal degradation patterns are dictated by other heterogeneous reactions. The different thermal degradation routes of varying heterogeneous reactions are reflected by different mass decay patterns in the TGA results. As observed in Fig. 4, the slopes or DTG values for the mass loss from Boudouard reaction are different. Besides any catalytic effects from CaCO₃ and CaO, this observation suggests that the kinetics of

Boudouard reaction is sensitive to the reaction substrates.

Overall, except for Boudouard reaction, CO_2 does not affect the physical aspects that govern the onset and end temperatures of thermolysis, thereby resulting in similar thermal degradation patterns. However, the chemical influence of CO_2 on thermolysis of each sample cannot be investigated through TGA.

3.3. Chemical influence of CO_2 on co-pyrolysis of chicken manure and biomass

To explore the chemical influence of CO_2 on the co-pyrolysis of chicken manure and biomass, scaled-up experiments for the pyrolysis of chicken manure, biomass, and the equivalent mass mixture of chicken manure and biomass were conducted separately under N_2 and CO_2 atmospheres. The concentration profiles of the major pyrolytic gases (H_2 , CH_4 , and CO) from the pyrolysis of chicken manure, biomass, and the equivalent mass mixture of both are shown in Figs. 5, 6 and 8, respectively.

Fig. 5 indicates the concentration profiles of the major pyrolytic gases as a function of the pyrolytic temperature from the pyrolysis of chicken manure under N_2 and CO_2 atmospheres. The gas evolution patterns under N_2 demonstrate a typical thermal degradation behavior of carbonaceous materials. As illustrated in Fig. 5(a), the formation of H_2 is expedited, proportional to the pyrolytic temperature. This phenomenon is likely attributed to thermal cracking followed by dehydrogenation. At 680°C , the concentration of H_2 reaches a maximum and subsequently declines. This phenomenon implies that the source of H_2 is fully depleted since the experimental setup is batch-type. The concentration profile of CH_4 also follows a typical evolution pattern for pyrolysis of carbonaceous materials. The origin of CH_4 can be explained by random bond scission. Considering that H_2 forms through dehydrogenation, the formation of CH_4 is observed in a lower temperature regime compared with that of H_2 . Hence, CH_4 can be used as a reaction substrate for forming H_2 via dehydrogenation. The formation of H_2 and CH_4 also provides a favorable condition for cyclization (charring compounds), resulting in the insignificant formation of CO in the high temperature regime. For instance, the formation of CO from the thermolysis of chicken manure was identified from 330°C

to 480°C as shown in Fig. 5(c). Interestingly, the formation of CO was identified above 660°C for the pyrolysis of chicken manure under N_2 . This observation indicates several features for the pyrolysis of chicken manure. Furthermore, CO_2 from CaCO_3 provides CO through Boudouard reaction. As discussed in sections 3.1 and 3.2, Boudouard reaction is thermodynamically favorable above 720°C . The TGA results in Figs. 1, 3 and 4 experimentally proved this. Thus, CaO generated from the dissociation of CaCO_3 into CO_2 seems to catalyze Boudouard reaction.

However, the concentration profiles under CO_2 deviate from typical pyrolysis pattern. As shown in Fig. 5(c), the generation of CO , which initiated above 550°C , is enhanced under CO_2 atmosphere. This suggests the effect of CO_2 on the pyrolysis of chicken manure as the initiation of Boudouard reaction was adjusted to 660°C despite the catalytic effects of CaO and/or CaCO_3 . Therefore, this phenomenon indicates an unknown reaction to form CO , which may not produce a catalytic effect based on different reaction temperatures. Furthermore, the amount of CH_4 generated from the pyrolysis of chicken manure under CO_2 is comparable with that under N_2 . Despite the similar evolution pattern for CH_4 , the formation of CH_4 under CO_2 initiated in a lower temperature regime compared with that under N_2 . This observation signifies that CO_2 expedites random bond scission during the pyrolysis of chicken manure. Moreover, as depicted in Fig. 5(b), the concentration of CH_4 starts to decrease at 550°C , where the enhanced generation of CO under CO_2 is initiated. This decrease is attributed to the dilution effect from the enhanced generation of CO under CO_2 because the GC only measures relative mole fraction. Hence, the relative low concentration of H_2 under CO_2 can be explained by this dilution effect. Overall, CO_2 functions as an oxygen donor to form CO and expediting agent for random bond scission in the pyrolysis of chicken manure. Moreover, although not catalytic, CO_2 reacts homogeneously with volatile hydrocarbons evolved from the pyrolysis of chicken manure. This interpretation is consistent with previous experimental findings displayed in Figs. 1, 3 and 4 because heterogeneous reactions are reflected in the TGA results.

For further investigation, similar experiments were conducted with biomass under N_2 and under CO_2 . Major pyrolytic gases were also quantified as a function of pyrolytic temperature as shown in

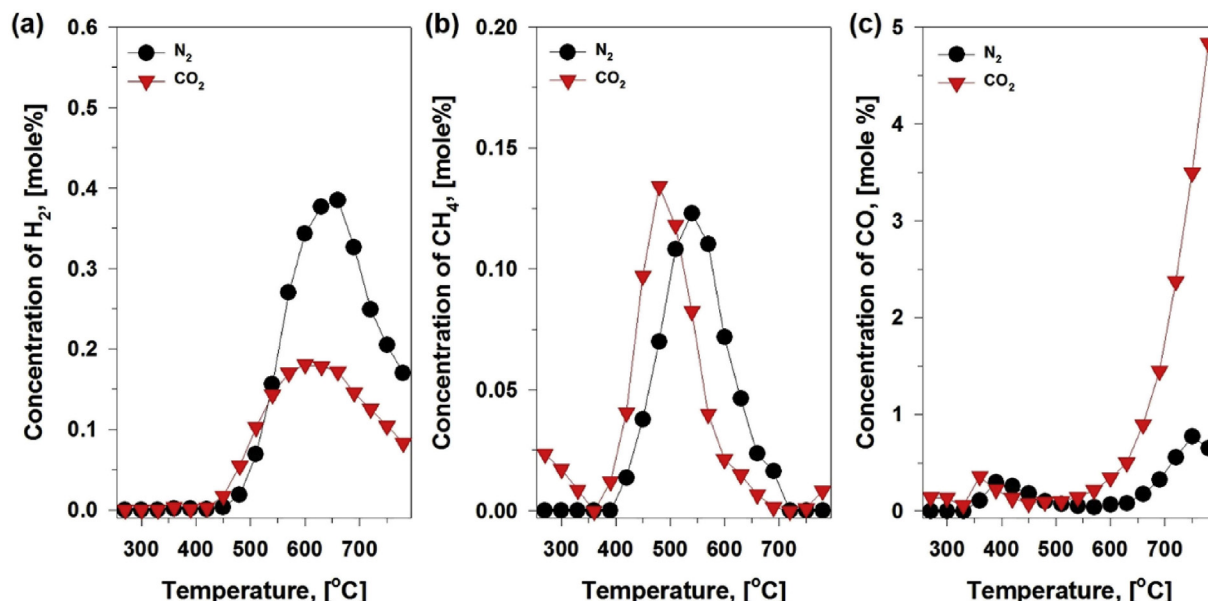


Fig. 5. Concentration profiles of (a) H_2 , (b) CH_4 , and (c) CO for the pyrolysis of chicken manure (1.4 ± 0.02 g) under N_2 and CO_2 (Error bar was not shown due to small deviation).

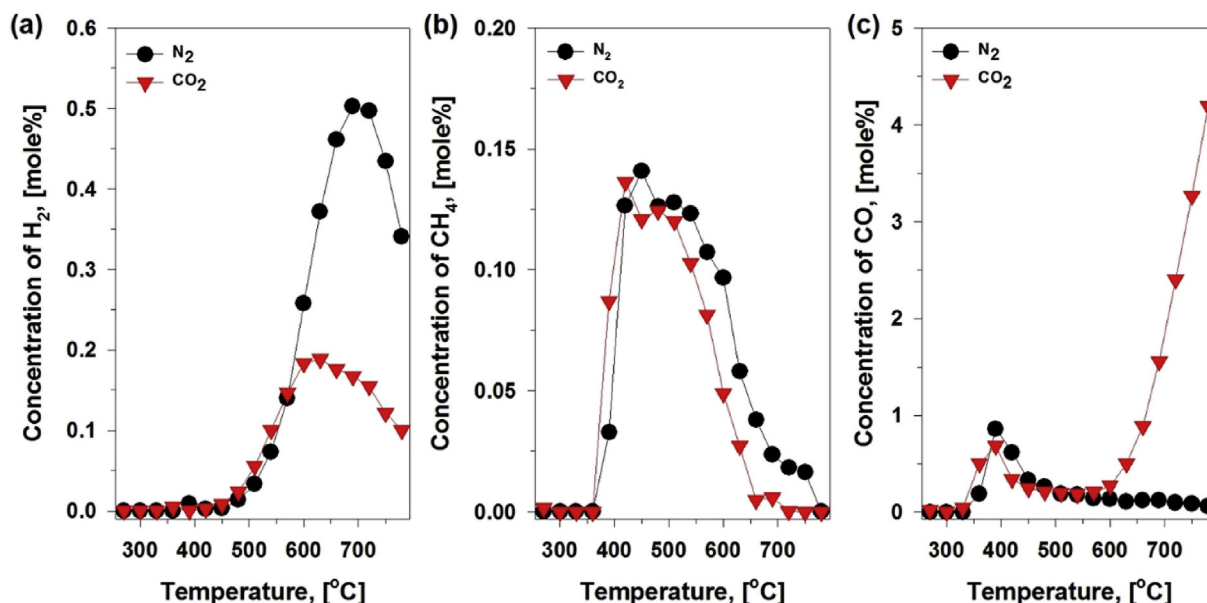


Fig. 6. Concentration profiles of (a) H_2 , (b) CH_4 , and (c) CO for the pyrolysis of biomass (1.4 ± 0.02 g) under N_2 and CO_2 (Error bar was not shown due to small deviation).

Fig. 6. Observably, the gas evolutions in the pyrolysis of biomass follow similar patterns with that in Fig. 5. Hence, CO_2 functions consistently in the pyrolysis of chicken manure and biomass. In earlier discussions, we claimed that the unknown reaction that forms CO is not a catalytic reaction. To verify this, the experimental data associated with the enhanced generation of CO from the pyrolysis of chicken manure and biomass in the presence of CO_2 were reprocessed. Fig. 7 evidently suggests that the gas evolutions of CO from both experiments have similar pattern. Furthermore, even in the presence of CaO and/or $CaCO_3$, the gas evolution of CO behaves consistently, further suggesting that the formation of CO is not a catalytic reaction. Moreover, the enhanced generation of CO in the presence of CO_2 occurred from 550 to 660 °C could not have been influenced by Boudouard reaction as this is observed only above 660 °C.

So far, the function of CO_2 in the pyrolysis of individual substrate (i.e., chicken manure and biomass) was observed and was confirmed to be universal. To achieve more realistic data, the co-

pyrolysis of the equivalent mass mixture of chicken manure and biomass was conducted under N_2 and CO_2 . The major pyrolytic gases are quantitatively plotted in Fig. 8.

The concentration profiles of the major pyrolytic gases (H_2 , CH_4 , and CO) from co-pyrolysis of chicken manure and biomass under N_2 and CO_2 atmospheres resemble those from the individual substrates (i.e., chicken manure and biomass). Hence, all experimental findings in Fig. 8 are consistent with earlier discussions.

3.4. Shifting carbon distribution from pyrolytic oil to syngas

In the previous sections, the function of CO_2 in the pyrolysis of carbonaceous materials was observed and its influence was selectively analyzed under different experimental temperature. CO_2 produces CO during the co-pyrolysis of chicken manure and biomass from 550 to 660 °C. This signifies that CO_2 provides a means for shifting carbon distribution from pyrolytic oil to syngas. The enhanced generation of syngas results in the subsequent reduction of pyrolytic oil as observed in the TGA results. The initial masses before Boudouard reaction under N_2 and CO_2 were the same. Therefore, CO_2 functions as both carbon scavenger and oxygen donor to form CO . To confirm this, the representative chromatograms of condensable hydrocarbons (i.e., pyrolytic oil) from the co-pyrolysis of chicken manure and biomass were obtained as shown in Fig. 9. The corresponding chemical species were qualitatively compared with their peak area in Table 2.

Fig. 9 and Table 2 suggest that the amount of chemical species in pyrolytic oil is substantially reduced in the presence of CO_2 , which supports the genuine role of CO_2 in the co-pyrolysis of chicken manure and biomass. Considering that peak area is proportional to concentration, Fig. 9 and Table 2 imply that all constituents in the presence of CO_2 were reduced, consistent with earlier discussions. CO_2 behaves both as a carbon scavenger and oxygen donor to form CO , shifting carbon distribution from pyrolytic oil to syngas. Thus, using CO_2 in pyrolysis generates more syngas even without a catalyst. Consequently, CO_2 can be applied in gasification process of carbonaceous materials for the same function because pyrolysis is an intermediate step. Since the formation of tar is a technical challenge in gasification, utilizing CO_2 as an initial feedstock can be

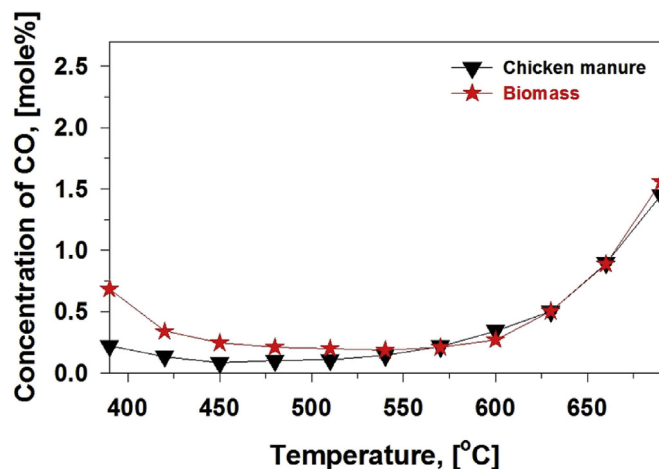


Fig. 7. Concentration profiles of CO for the pyrolysis of chicken manure (1.4 ± 0.02 g) and biomass (1.4 ± 0.02 g) in CO_2 (Error bar was not shown due to small deviation).

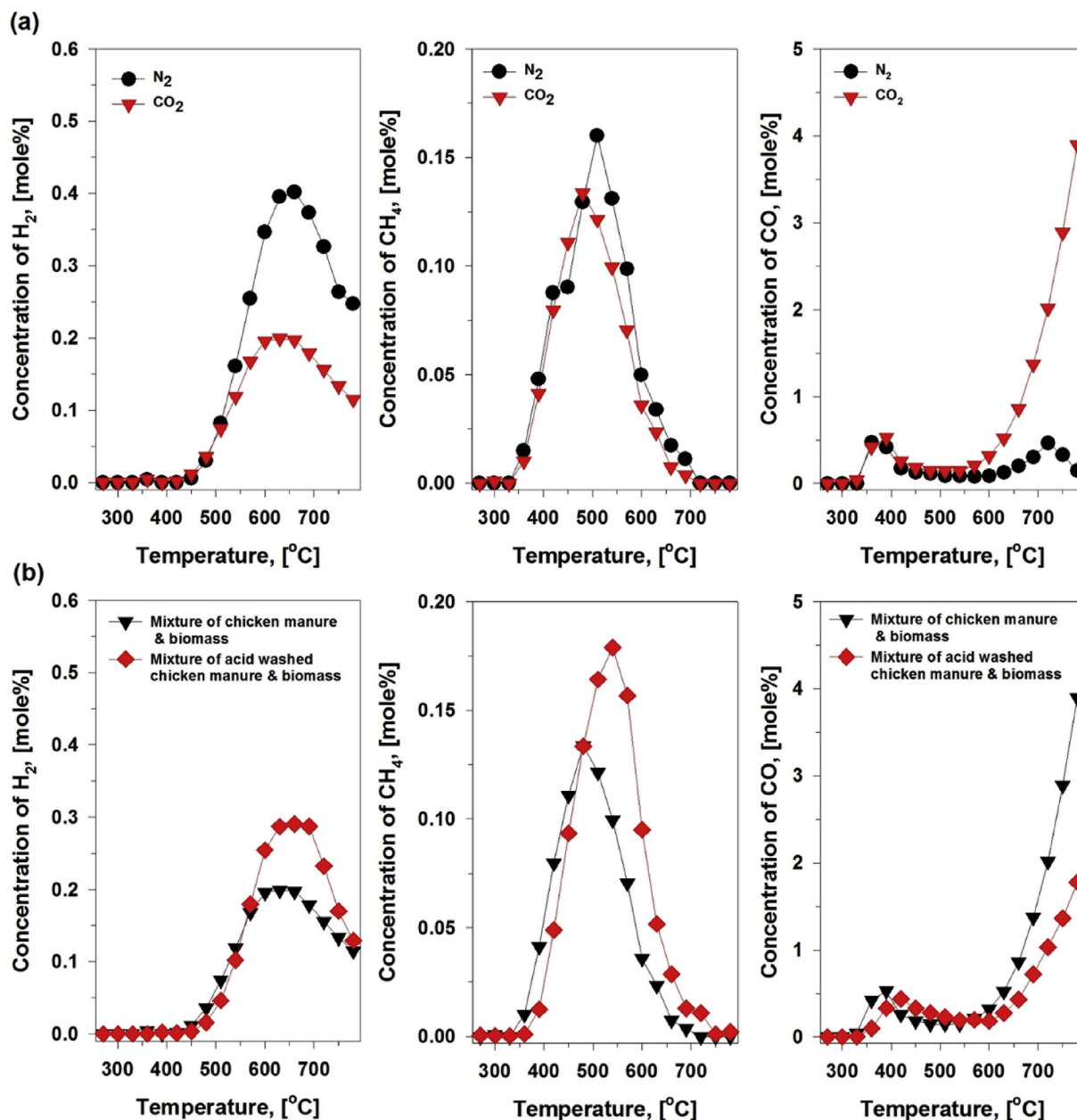


Fig. 8. (a) Concentration profiles of H_2 , CH_4 , and CO for the co-pyrolysis of chicken manure and biomass (1.4 ± 0.02 g) under N_2 and CO_2 . (b) Comparison of the gas evolutions from the co-pyrolysis of acid-washed chicken manure and biomass and chicken manure and biomass under CO_2 only (Error bar was not shown due to small deviation).

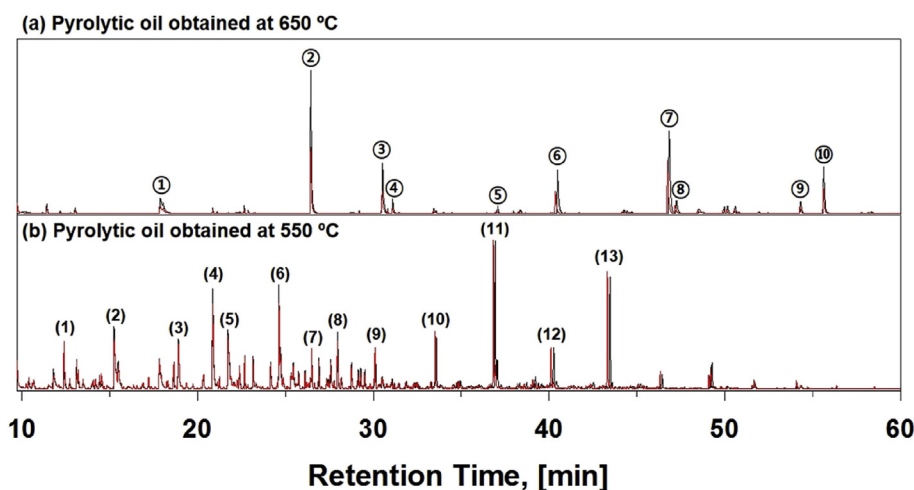


Fig. 9. Representative chromatograms for pyrolytic oil from the co-pyrolysis of the equivalent mass mixture of chicken manure and biomass in N_2 (black color) and CO_2 (red color). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 2Summary of the major chemical species in pyrolytic oil from the co-pyrolysis of the equivalent mass mixture of chicken manure and biomass under N₂ and CO₂.

Label number	Retention Time, [min]	Chemical species	Peak Area in N ₂	Peak Area in CO ₂	Peak Area difference, [%]
①	17.8726	Phenol	244051000	57666500	−76.37
②	26.4411	Naphthalene	680310000	161707000	−76.23
③	30.5237	Methylnaphthalene	297129000	65873000	−77.83
④	31.1034	2-Methylnaphthalene	71420600	15007425	−78.99
⑤	37.0845	Acenaphthene	43007300	9004775	−79.06
⑥	40.4951	9H-Fluorene	271757000	71036250	−73.86
⑦	46.8459	Phenanthrene	533618000	169089000	−68.31
⑧	47.2659	3,4,5,6,7,8-Hexahydro-1,1-dimethyl-2(1H)-naphthalenone	99593800	30761750	−69.11
⑨	50.6009	4H-Cyclopenta[Def]phenanthrene	43740200	11672750	−73.31
⑩	55.6496	Pyrene	279184000	24262825	−91.31
(1)	12.3805	1-Propene	207564000	229427438	10.53
(2)	15.2451	3,6,7-Trimethyl-triazolo[4,3-B]-1,2,4-triazine	732985000	446985563	−39.02
(3)	18.9077	2,2-Dimethyl-4-oxo-3-oxabicyclo[3.1.0]hexane-5,6,6-tricarbonitrile	482959000	332341313	−31.19
(4)	20.8566	2-Methylphenol	806531000	524283750	−35.00
(5)	21.7219	4-Methylphenol	780972000	512822250	−34.34
(6)	24.6285	3,5-Dimethylphenol	940627000	593803125	−36.87
(7)	26.5018	Dodecane	159428000	200724750	25.90
(8)	27.9719	2-Ethyl-4-methylphenol	366351000	236301750	−35.50
(9)	30.1224	Tridecane	200059000	177922688	−11.06
(10)	33.575	Tetradecane	344459000	259666875	−24.62
(11)	36.9436	Pentadecane	796456000	580404375	−27.13
(12)	40.2702	Hexadecane	222890000	159725813	−28.34
(13)	43.4708	Heptadecane	588436000	464375813	−21.08

promising and environmentally beneficial. All experimental findings in this study were not fully optimized. Hence, further study is required to maximize the identified role of CO₂ in the thermochemical process and to apply it in gasification process.

4. Conclusions

The role of CO₂ in the co-pyrolysis of chicken manure and biomass was mainly investigated in this study. TGA confirmed that the physical influence of CO₂ was nearly negligible. However, the high content (17 wt%) of CaCO₃ in chicken manure catalyzed Boudouard reaction. This study also revealed the chemical influence of CO₂ on co-pyrolysis of chicken manure and biomass, manifested by expedited thermal cracking and formation of CO. The enhanced generation of CO in the presence of CO₂ suggests that CO₂ acted as an oxygen donor, which shifted carbon distribution between pyrolytic products.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.energy.2018.04.084>.

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